



Conference Abstract

Nutrient input to large rivers: From Monitoring to Modelling localized groundwater discharge and its effects on eutrophication in 1st order river systems

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Abstract

In cultural landscapes as in Central Europe, groundwater baseflow is often strongly enriched in nutrients (N; P) and reaches surface water bodies. Once approached, nutrients may lead to eutrophication within the receiving surface water and eventually in the ocean. Any intervening measures must rely on a fundamental knowledge about the amount of baseflow, the regions where it enters the water bodies and what is the ecological hazard resulting from it. Although groundwater usually differs chemically and isotopically from gaining surface water bodies, particularly in 1st order rivers like the Elbe, groundwater inflow is camouflaged due to the high dilution. Nevertheless, the present study addresses the spatial, temporal and volumetric determination and dynamics of groundwater discharge, its attributed nutrient contribution and the systematic

analysis of its impact on benthic and planktonic eutrophication to the river Elbe. Therefore, water samples were taken from:

- 1. the Elbe every 2 km along a 450 km long reach of the river,
- 2. the inflowing streams, and
- 3. wells and springs, representing the groundwater bodies along the river.

All samples have been analysed for ionic composition, water isotopes nad nutrients. In a first step and on the base of:

- 1. analysis of daily time series of hydraulic gradients between river- and groundwater levels,
- 2. flux balances for river segments,
- 3. inverse geochemical modelling of the river water composition,
- 4. a Darcy approach based on the hydraulic conductivity of the upper aquifer and
- 5. a model of the tritium dilution effect along the river water course, we were able to identify locations and the temporal dynamics of the interaction of groundwater with the river Elbe (Fig. 1).

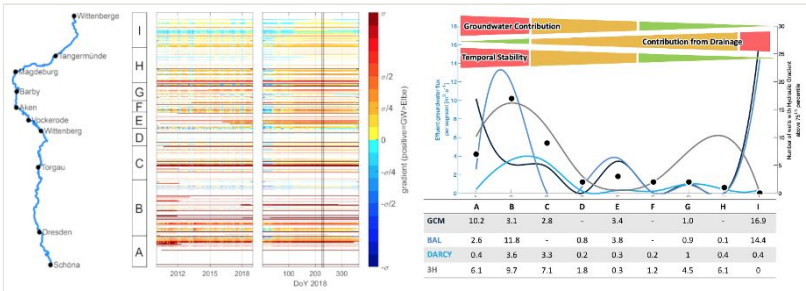


Figure 1. [doi](#)

Fig. 1: A) shows investigated reach of the Elbe, being divided into sub-reaches A-I; central panel: normalised hydraulic gradients for years 2010 - 2018 and for the year 2018 (DoY=days of the year).

B) Combined results of qualitative and quantitative approaches regarding groundwater contributions, temporal stability and the deduced contribution from field drainages based (blue ordinate axis refers to all line elements; black ordinate axis refers to point elements in the figure).

The groundwater samples were additionally analysed for radiocarbon, $^3\text{He}/^4\text{He}$ and anthropogenic gases (SF_6 , CFCs) to determine their average age, being important when it comes to management strategies to reduce the potentially harmful function of the groundwater. Since fertilization in agriculture is the major reason for the high nutrient load in groundwaters, to intervene it is important to know, whether the nitrate approaching the river stems from the current farming activities or from decades ago. Our results showed, groundwaters approaching the river are usually younger than 40 years, indicating modern regulatory measures such as German Fertilizer Regulation and the EU-WFD may have an impact on the nutrient load of the Elbe.

To estimate how these groundwater-borne nutrient inflows affect the eutrophication of the river Elbe, we quantified the input and determined the impacts on local (benthic biofilms) and regional (phytoplankton) eutrophication in areas along the river, by monitoring biofilm growth on exposed artificial substratum in river segments and the abundance and composition of phytoplankton. The experimental benthos setup included sites where groundwater inflow was proven and others without proven groundwater inflow. Further, we differentiated between the types of aquifers and their hydraulic connection: unconsolidated Quaternary aquifers, which are supposed to let groundwater pass through the pore space and pre-Neogene hard rock aquifers, considered to restrict groundwater flow to faults and cracks. To also understand possible seasonal effects, the in-situ monitoring was repeated during springtime, summer and autumn, while effects on pelagic biomass production was monitored during summer only.

As a result, along the 450 km stretch, groundwater contributes under low flow conditions daily up to 1.5 t PO₄, which enter the river diffusely and significantly enhances the eutrophication risk, since the availability of phosphorus usually limits biomass production. Groundwater-P may hence result in an additional daily planktonic load of about 46 tons of particulate organic carbon, thereby contributing to eutrophication at the regional scale in the Elbe River. At the local benthic scale, analyses of biofilm community composition, biofilm macronutrients, and structural components react distinctly on the availability of groundwater and showed seasonal effects. Wherever groundwater inflow is high, as in regions dominated by unconsolidated, highly conductive Quaternary aquifers, benthic eutrophication takes place and most likely during autumn. The strong interaction of environmental factors in determining benthic eutrophication highlights the need to assess these factors in combination rather than in isolation.

Our assessment of a large European 1st order river with high eutrophication risk for the river itself and the German Bay provides the status-quo only and calls for systematic long-term monitoring of ground- and surface water composition as well as ecological indicators to evaluate the success of regulatory measures in respect to ecological quality in that complex and interacting system.

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Conflicts of interest

The authors have declared that no competing interests exist.